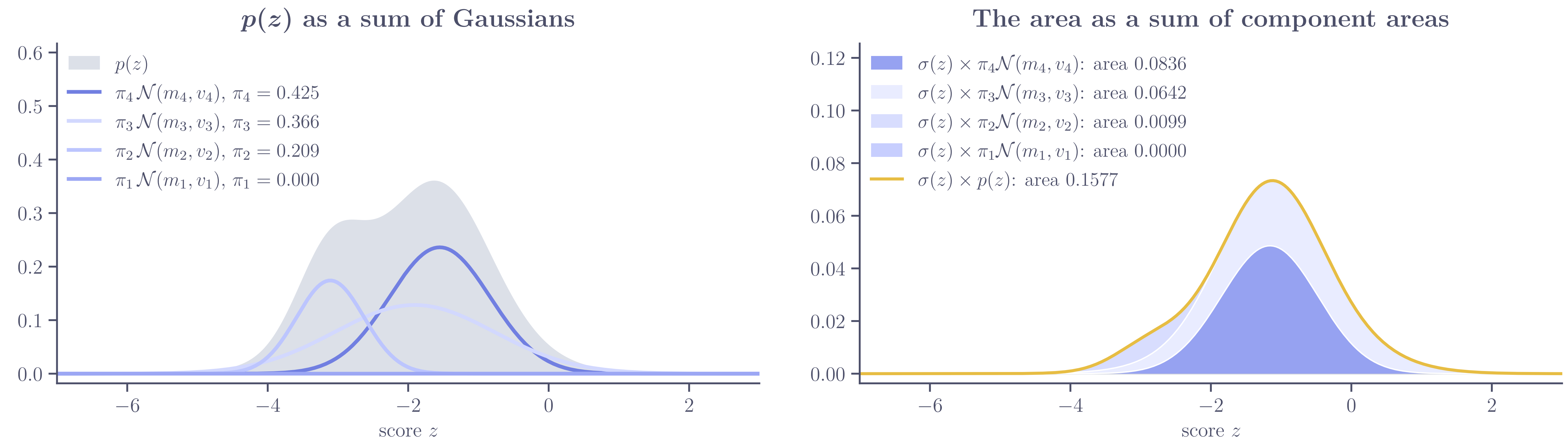
STEP 1

$p(z)$ is a mixture of Gaussians, so the area splits by component

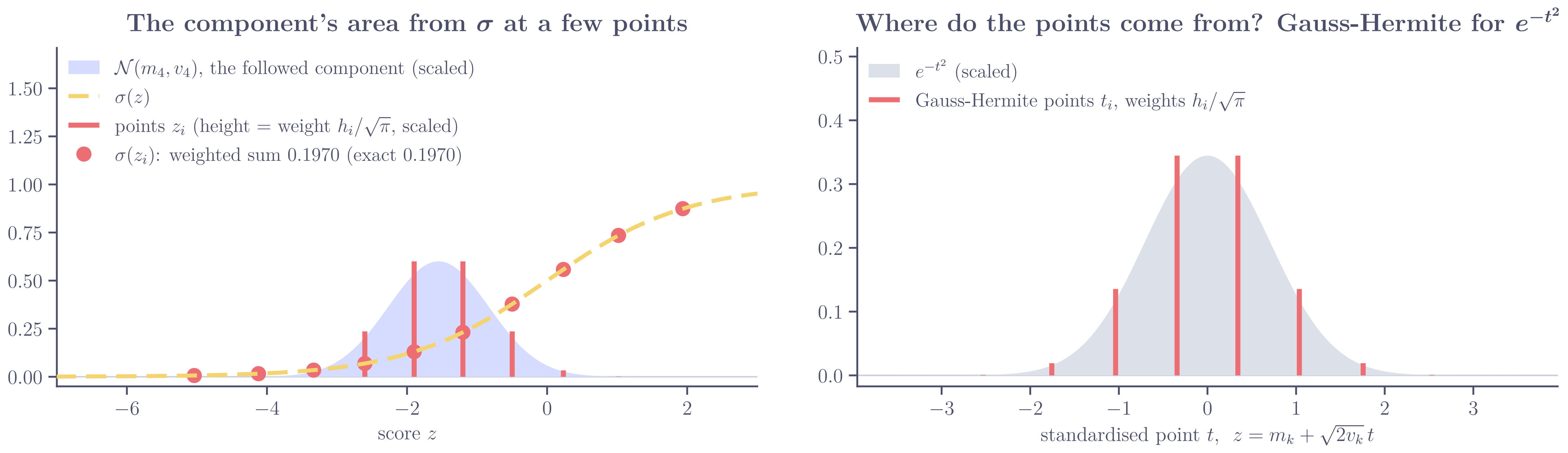

$p(z)$ as a sum of Gaussians

The area as a sum of component areas

For a Gaussian mixture model the goal of the PCPG step by step, the area under $\sigma(z) \times p(z)$, can be computed directly, with one small quadrature per component, because $p(z)$ itself is known. Given the observed features, z is again a mixture: in component k it is Gaussian, $z \mid k \sim \mathcal{N}(m_k, v_k)$, with weight $\pi_k = p(k \mid x_{\text{obs}})$, mean $m_k = w^\top \mu_k$ (observed features filled in) and variance $v_k = \sum_{j\,\text{missing}} w_j^2 \sigma_{kj}^2$. So the area splits into one area per component: $\mathbb{E}[\sigma(z)] = \sum_k \pi_k\, \mathbb{E}[\sigma(z) \mid k]$, and each $\mathbb{E}[\sigma(z) \mid k]$ is a 1D integral of σ against a Gaussian. The dark indigo component (largest weight) is followed in the next steps.

STEP 2

The area of a component is a weighted sum of $\boldsymbol{\sigma}$ at a few points

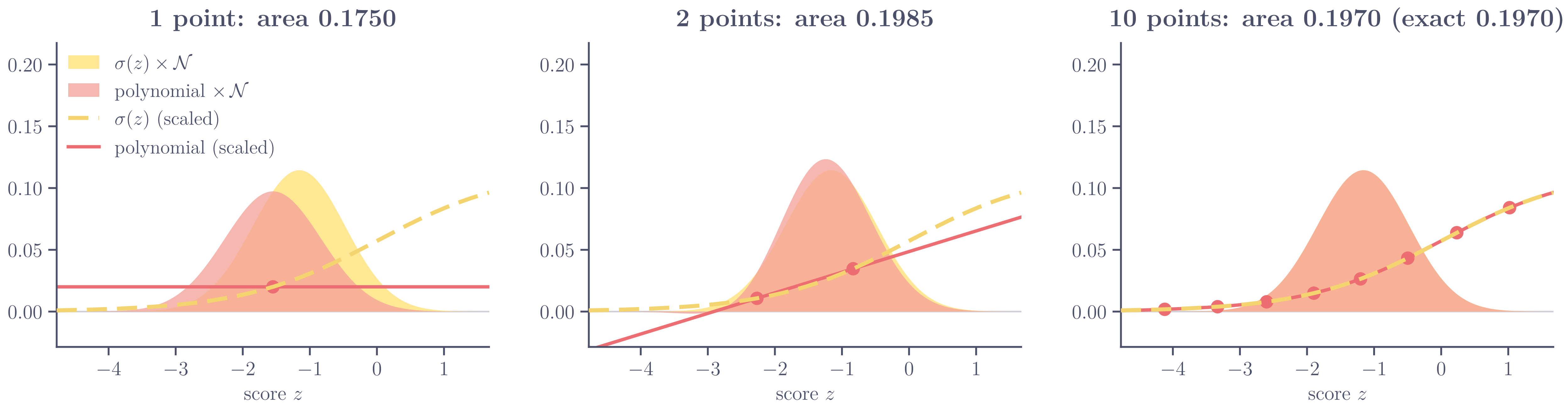

The component's area from σ at a few points

Where do the points come from? Gauss-Hermite for e^{-t^2}

Gauss-Hermite replaces the integral of $\sigma(z)$ against the Gaussian of the component by a weighted sum $\sum_i \frac{h_i}{\sqrt{\pi}} \sigma(z_i)$ over 10 points z_i (lines, height = weight). Only σ is evaluated, the Gaussian enters through where the points are and how much they weigh. The points t_i and weights h_i are fixed once, for the weight e^{-t^2} (grey), and moved to every component by $z_i = m_k + \sqrt{2v_k}\, t_i$. They are chosen so that the sum is exact for every polynomial up to degree $2n - 1$: the points crowd where the Gaussian has its mass.

STEP 3

Why it works: the sum is the area under a polynomial through $\boldsymbol{\sigma}$


1 point: area 0.1750

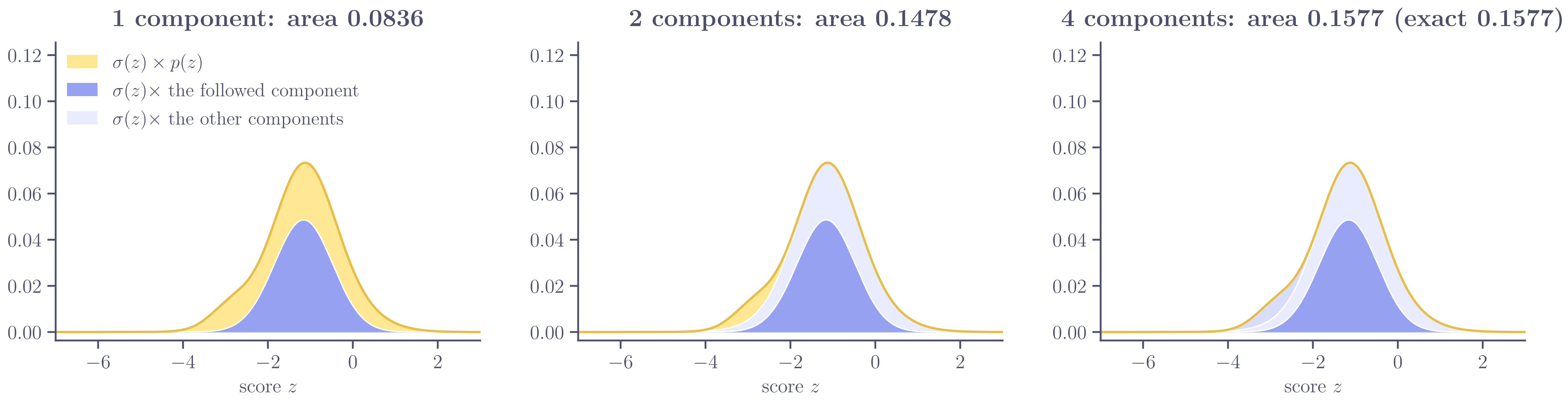
2 points: area 0.1985

10 points: area 0.1970 (exact 0.1970)

Every panel: with n points the sum is exactly the area under the polynomial of degree $n - 1$ through σ at the points (coral line and dots, scaled) times the Gaussian of the component (coral area), against $\sigma(z)$ (yellow, dashed) and its area. One point is a constant, two a line. As the polynomial closes in on σ where the Gaussian has its mass, the area closes in on the component's area. Far out the polynomial runs off, but there the Gaussian is about 0.

STEP 4

The component areas add up to the area under $\boldsymbol{\sigma(z) \times p(z)}$


1 component: area 0.0836

2 components: area 0.1478

4 components: area 0.1577 (exact 0.1577)

Every panel: $\sigma(z)$ times the sum of the components with the largest weights (dark indigo, then lighter, stacked) and its area $\sum_k \pi_k\, \mathbb{E}[\sigma(z) \mid k]$, every component area from 10 Gauss-Hermite points, against the goal (yellow). With 4 components it is the GMM Gauss-Hermite estimate 0.1577 (exact 0.1577).